

19 March 2019

①

~moller/doc/runtemplate\_2019\_1.pdf

| Date     | Begin time | End time | Crew          | Comments             |
|----------|------------|----------|---------------|----------------------|
| 03/19/19 | 12:35      |          | Simona, Roman | Hel 30 Hz, Delay = 8 |

| Energy | Linac                | Injector | Pass | Wien Angle | HWP | Slit | Attenuat | Raster |
|--------|----------------------|----------|------|------------|-----|------|----------|--------|
| 2.138  | $E=1000$<br>$N=1020$ | 119      | 1    | 60         | OUT | 20.7 | 240      | OFF    |

|      | Q1           | Q2          | Q3 | Q4        | Dipole        |
|------|--------------|-------------|----|-----------|---------------|
| Amps | 100.37/2.5 G | 23.57/0.5 G | 0  | 41.37/1.6 | 106.39/2.88 G |

| N  | Time  | I <sub>beam</sub><br>Curr | Run<br>Number                             | Targ<br>et | H <sub>coil</sub><br>Field | Polarization<br>% | FADC       | Comments                      |
|----|-------|---------------------------|-------------------------------------------|------------|----------------------------|-------------------|------------|-------------------------------|
| 1  | 12:35 |                           | ask MCC for Moller merge settings         |            |                            |                   |            |                               |
| 2  | 13:25 |                           | start ramping Solenoid to 2.5 T (46.64 A) |            |                            |                   |            |                               |
| 3  | 13:58 |                           | Solenoid set to 2.5 T                     |            |                            |                   |            |                               |
| 4  | 14:00 |                           | ask MCC to check beam orbit               |            |                            |                   |            |                               |
| 5  | 14:17 |                           | No beam for next ~20 min                  |            |                            |                   |            |                               |
| 6  | 14:52 |                           | Set target #2 (10 $\mu$ m)                |            |                            |                   |            |                               |
| 7  |       |                           |                                           |            |                            |                   |            |                               |
| 8  | 15:05 | 0.3                       | 16290                                     | 2          | 2.5T                       |                   | 130 kHz    | Thresh check = 180 No<br>6ea. |
| 9  | 15:10 | 0.3                       | 16291                                     | 2          | 2.5T                       |                   | 45 kHz     | Thresh = 180 R+rig            |
| 10 | 15:13 | 0.3                       | 16292                                     | 2          | 2.5T                       |                   | 43 kHz     | Thresh = 200                  |
| 11 | 15:17 | 0.3                       | 16293                                     | 2          | 2.5T                       |                   | 40 kHz     | Thresh = 230                  |
| 12 | 15:21 | 0.3                       | 16294                                     | 2          | 2.5T                       |                   |            | Thresh = 250                  |
| 13 | 15:25 | 0.3                       | 16295                                     | 2          | 2.5T                       |                   |            | Thresh = 230 L+rig            |
| 14 |       |                           |                                           |            | Target #2 (10 $\mu$ m)     |                   | Angle scan |                               |
| 15 | 15:30 | 0.3                       | 19296                                     | 2          | 2.5T                       | 77.49 $\pm 0.53$  |            | Angle = -6.48                 |
| 16 | 15:41 | 0.3                       | 19297                                     | 2          | 2.5T                       | 79.47 $\pm 0.48$  |            | Angle = -4.482                |
| 17 | 15:49 | 0.3                       | 19298                                     | 2          | 2.5T                       | 83.3 $\pm 0.5$    |            | Angle = -2.484                |
| 18 | 15:58 | 0.3                       | 19299                                     | 2          | 2.5T                       | 83.12 $\pm 0.49$  |            | Angle = 0°                    |
| 19 | 16:05 | 0.3                       | 19300                                     | 2          | 2.5T                       | 81.99 $\pm 0.48$  |            | Angle = +1.998                |
| 20 | 16:15 | 0.3                       | 19301                                     | 2          | 2.5T                       | 79.41 $\pm 0.48$  |            | Angle = +3.996                |
| 21 | 16:26 | 0.3                       | 19302                                     | 2          | 2.5T                       | 76.95 $\pm 0.48$  |            | Angle = +6.840                |
| 22 | 16:35 | 0.3                       | 19303                                     | 2          | 2.5T                       | 78.5 $\pm 0.48$   |            | Angle = +4.842                |
| 23 | 16:43 | 0.3                       | 19304                                     | 2          | 2.5T                       | 78.22 $\pm 0.49$  |            | Angle = +3.996                |
| 24 | 16:51 | 0.3                       | 19305                                     | 2          | 2.5T                       | 84.82 $\pm 0.49$  |            | Angle = 0                     |
| 25 | 17:00 | 0.3                       | 19306                                     | 2          | 2.5T                       | 82.94 $\pm 0.52$  |            | Angle = -2.484                |
| 26 | 17:11 | 0.3                       | 19307                                     | 2          | 2.5T                       | 77.63 $\pm 0.48$  |            | Angle = -6.480                |



| N  | Time             | I <sub>beam</sub><br>curr | Run<br>Number | Targ<br>et                    | H <sub>coil</sub><br>Field | Polarization<br>% | FADC | Comments       |
|----|------------------|---------------------------|---------------|-------------------------------|----------------------------|-------------------|------|----------------|
| 1  | 17:21            | 0.3                       | 16308         | 2                             | 2.5T                       | 76.87 ± 0.49      |      | Angle = 6.840  |
| 2  | 17:31            | 0.3                       | 16309         | 2                             | 2.5T                       | 78.94 ± 0.60      |      | Angle = -4.482 |
| 3  | 17:43            | 0.3                       | 16310         | 2                             | 2.5T                       | 84.93 ± 0.50      |      | Angle = 0°     |
| 4  | 17:53            | 0.3                       | 16311         | 2                             | 2.5T                       | 80.91 ± 0.48      |      | Angle = 1.998  |
| 5  |                  |                           |               | Set Target #3 (4 μm)          |                            |                   |      |                |
| 6  | 18:03            | 0.                        | 16312         |                               | target 2 → #3              |                   |      |                |
| 7  | 18:06            | 0.5                       | 16313         | 3                             | 2.5T                       | 80.37 ± 0.54      |      | Angle = 1.998  |
| 8  | 18:16            | 0.5                       | 16314         | 3                             | 2.5T                       | 84.0 ± 0.54       |      | Angle = 0      |
| 9  | 18:26            | 0.5                       | 16315         | 3                             | 2.5T                       | 83.35 ± 0.52      |      | Angle = -1.998 |
| 10 | 18:33            | 0.5                       | 16316         | 3                             | 2.5T                       | 80.48 ± 0.52      |      | Angle = -6.48  |
| 11 | 18:43            | 0.5                       | 16317         | 3                             | 2.5T                       | 76.06 ± 0.54      |      | Angle = +6.840 |
| 12 |                  |                           |               | Ramping Solenoid to 3T        |                            |                   |      |                |
| 13 | 19:04            |                           |               | Cycling Dipole to 125.84 Amps |                            |                   |      |                |
| 14 | 19:07            | 0.5                       | 16318         | 3                             | 3T                         | 21.68 kHz         | coin | 125.84         |
| 15 | 19:13            | 0.5                       | 16319         | 3                             | 3T                         | 24.15 kHz         |      | 122.364        |
| 16 | 19:18            | 0.5                       | 16320         | 3                             | 3T                         | 28.199 kHz        |      | 111.717        |
| 17 | 19:23            | 0.5                       | 16321         | 3                             | 3T                         | 30.23 kHz         |      | 106.384        |
| 18 | 19:              | 0.5                       | 16322         | 3                             | 3T                         | 29.79 kHz         |      | 101.07         |
| 19 | 19:33            | 0.5                       | 16324         | 3                             | 3T                         | 26.76 kHz         |      | 95.747         |
| 20 | 19:39            | 0.5                       | 16325         | 3                             | 3T                         | 19.9 kHz          |      | 90.424         |
| 21 |                  |                           | — Q1 Scan —   |                               |                            |                   |      |                |
| 22 | 20:25            | 0.5                       | 16326         | 3                             | 3T                         | 30.913 kHz        |      | 115.5 A.       |
| 23 | <del>20:40</del> | 0.5                       | 16327         | 3                             | 3T                         | 31.237 kHz        |      | 115.5 A.       |
| 24 | 20:40            | 0.5                       | 16328         | 3                             | 3T                         | 32.17 kHz         |      | 110.453 A.     |
| 25 |                  | 0.5                       | 16329         | 3                             | 3T                         | 31.56 kHz         |      | 110.453 A.     |
| 26 | 21:05            | 0.5                       | 16330         | 3                             | 3T                         | 32.93 kHz         |      | 105.411 A.     |
| 27 |                  | 0.5                       | 16331         | 3                             | 3T                         | 33.75 kHz         |      | 105.411 A.     |
| 28 |                  | 0.5                       | 16332         | 3                             | 3T                         | 33.88 kHz         |      | 105.411 A.     |
| 29 |                  | 0.5                       | 16333         | 3                             | 3T                         | 33.23 kHz         |      | 105.411 A.     |
| 30 | <del>21:40</del> | <del>0.5</del>            |               |                               |                            |                   |      |                |



| N    | Time  | I <sub>beam</sub><br>curr | Run<br>Number | Targ<br>et | H <sub>coil</sub><br>Field | Polarization<br>% | FADC | Comments  |
|------|-------|---------------------------|---------------|------------|----------------------------|-------------------|------|-----------|
| 1    |       |                           |               |            |                            |                   |      |           |
| 2    | 21:45 | 0.5                       | 16334         | 3          | 3T                         | 34.07 KHz         |      | 95.334 A. |
| 3    |       |                           | 16335         | 3          | 3T                         | 34.23 KHz         |      | 95.334 A. |
| 4    |       |                           | 16336         | 3          | 3T                         | 33.24 KHz         |      | 95.334 A. |
| 5    |       |                           | 16337         | 3          | 3T                         | 33.91 KHz         |      | 95.334 A. |
| 6    | 22:25 | 0.5                       | 16338         | 3          | 3T                         | 29.518 KHz        |      | 90.298 A. |
| 7    |       |                           | 16339         | 3          | 3T                         | 29.757 KHz        |      | 90.298 A. |
| 8    |       |                           | 16340         | 3          | 3T                         | 29.957 KHz        |      | 90.298 A. |
| 9    | 22:55 | 0.5                       | 16341         | 3          | 3T                         | 25.53 KHz         |      | 85.262 A. |
| 10   |       |                           | 16342         | 3          | 3T                         | 25.168 KHz        |      | 85.262 A. |
| 11   |       |                           | 16343         | 3          | 3T                         | 24.41 KHz         |      | 85.262 A. |
| 12   |       |                           | 16344         | 3          | 3T                         | 26.09 KHz         |      | 85.262 A. |
| 13   |       |                           |               |            |                            |                   |      |           |
| 14   |       |                           |               |            |                            | Q2 Scans          |      |           |
| 15   | 23:40 | 0.5                       |               | 3          | 3T                         |                   |      | 38 A      |
| 16   |       | 0.5                       |               | 3          | 3T                         |                   |      | 38 A      |
| 17   |       |                           |               | 3          | 3T                         |                   |      |           |
| 18   |       |                           |               |            |                            |                   |      |           |
| 19   |       |                           |               |            |                            |                   |      |           |
| 20   |       |                           |               |            |                            |                   |      |           |
| 21   | 00:42 | 0.5                       | 16348         | 3          | 3T                         |                   |      | 30.964 A  |
| 22   | 00:56 | 0.5                       | 16349         | 3          | 3T                         | 85.41 ± 0.44      |      | 16.172    |
| 23   | 01:08 | 0.5                       | 16350         | 3          | 3T                         | 86.50 ± 0.46      |      | 12.719    |
| 24   | 01:20 | 0.6                       | 16351         | 3          | 3T                         | 84.91 ± 0.51      |      | 10.745    |
| 25   | 1:33  |                           |               |            |                            | Set Q2            | →    | 23.57 A   |
| 26   |       |                           |               |            |                            | (cycle)           |      |           |
| 27   |       |                           |               |            |                            | Q4 scans (cycle)  |      |           |
| 28   | 1:41  | 0.5                       | 16352         | 3          | 3T                         | 86.29 ± 0.56      |      | 65.0      |
| 29   | 1:54  | 0.5                       | 16353         | 3          | 3T                         | 80.84 ± 0.45      |      | 53.807    |
| 30   | 2:04  | 0.5                       | 16354         | 3          | 3T                         | 83.05 ± 0.43      |      | 41.37     |
| * 31 | 2:14  | 0.5                       | 16355         | 3          | 3T                         | 87.67 ± 0.45      |      | 33.088    |

↑  
check Azz



Q4 scan (cont.)

| N  | Time  | I <sub>beam</sub><br>curr | Run<br>Number | Targ<br>et           | H <sub>coil</sub><br>Field | Polarization<br>% | FADC | Comments                                             |
|----|-------|---------------------------|---------------|----------------------|----------------------------|-------------------|------|------------------------------------------------------|
| 1  | 2:25  | 0,5                       | 16356         | 3                    | 3T                         | 87,09 0,46        |      | 31.018 A Q4                                          |
| 2  | 2:35  | 0,5                       | 16357         | 3                    | 3T                         | 89,61 0,53        |      | 28.948                                               |
| 3  |       |                           |               |                      |                            |                   |      |                                                      |
| 4  |       |                           |               | One Q2 solution      |                            |                   |      |                                                      |
| 5  |       |                           |               |                      |                            |                   |      | Q <sub>1</sub> , Q <sub>3</sub> , Q <sub>4</sub> = ∅ |
| 6  |       |                           |               |                      | Q2 scan                    |                   |      | D = 106,394                                          |
| 7  | 3:05  | 0,5                       | 16358         | 3                    | 3T                         | 82,85 ± 0,51      |      | Q <sub>2</sub> = 121,853 A                           |
| 8  | 3:14  | 0,5                       | 16359         | 3                    | 3T                         | 82,92 ± 0,5       |      | 121,853 A                                            |
| 9  | 3:30  | 0,5                       | 16360         | 3                    | 3T                         | 83,84 ± 0,77      |      | 140,257 A cycled                                     |
| 10 | 3:41  | 0,5                       | 16361         | 3                    | 3T                         | 82,12 ± 0,63      |      | 134,119                                              |
| 11 | 3:52  | 0,5                       | 16362         | 3                    | 3T                         | 83,76 ± 0,56      |      | 127,985                                              |
| 12 | 4:03  | 0,5                       | 16363         | 3                    | 3T                         | 83,82 ± 0,47      |      | 115,722 A <sub>22</sub>                              |
| 13 | 04:13 | 0,5                       | 16364         | 3                    | 3T                         | 83,14 ± 0,46      |      | 112,044 (0,7427)                                     |
| 14 | 04:24 | 0,5                       | 16365         | 3                    | 3T                         | 85,89 ± 0,50      |      | 110,572 (0,7219)                                     |
| 15 | 4:36  | 0,5                       | 16366         | 3                    | 3T                         | 87,51 ± 0,49      |      | 109,592                                              |
| 16 |       |                           |               | Set 3 Quads solution |                            |                   |      |                                                      |
| 17 |       |                           |               |                      |                            |                   |      | Q <sub>1</sub> = 100,371                             |
| 18 |       |                           |               |                      |                            |                   |      | Q <sub>2</sub> = 23,57                               |
| 19 |       |                           |               |                      |                            |                   |      | Q <sub>3</sub> = 0                                   |
| 20 |       |                           |               |                      | Q4 scan                    |                   |      | Q <sub>4</sub> = 41,37                               |
| 21 | 4:55  | 0,5                       | 16367         | 3                    | 3T                         | 84,13 ± 0,46      |      | 41,37                                                |
| 22 | 05:05 | 0,5                       | 16368         | 3                    | 3T                         | 27,64 kHz         |      | 30,                                                  |
| 23 | 05:10 | 0,5                       | 16369         | 3                    | 3T                         | 21,4 kHz          |      | 28,                                                  |
| 24 | 05:14 | 0,5                       | 16370         | 3                    | 3T                         | 15,72 kHz         |      | 26,                                                  |
| 25 | 05:17 | 0,5                       | 16371         | 3                    | 3T                         | 11,745 kHz        |      | 24,                                                  |
| 26 | 05:21 | 0,5                       | 16372         | 3                    | 3T                         | 7,71 kHz          |      | 22,                                                  |
| 27 | 5:25  | 0,5                       | 16373         | 3                    | 3T                         | 5,09 kHz          |      | 20,                                                  |
| 28 | 5:29  | 0,5                       | 16374         | 3                    | 3T                         | 3,07 kHz          |      | 18,                                                  |
| 29 | 5:43  | 0,5                       | 16375         | 3                    | 3T                         | 83,80 ± 0,63      |      | Q <sub>4</sub> = 41,37 (cycl.)                       |
| 30 | 5:    | 0,5                       | 16376         | 3                    | 3T                         | test_LED          | 0,   | Dead Time Test                                       |
|    | 6:03  | 0,5                       | 16377         | 3                    | 3T                         | 0,17<br>0,13 %    |      |                                                      |

6:08 1,0 16378 3 3T 0,27 %  
 6:21 1,1 16379 3 3T 0,29  
 0,3 16380 3 3T 0,1 %  
 1,2 16381-16384 0,31 %